

Homework 1

Solution

1. (a) Let $t \in \mathbb{R}$ and consider the linear system in the variables x, y, z given by

$$\begin{cases} x + 2y + 3z = 6 \\ 2x + 5y + 2z = 4 + t \\ 6x - 3y + z = 2 \end{cases}$$

Find the (unique) solution using Cramer's rule.

Solution:

The system can be written in matrix form as $A\vec{x} = \vec{b}$, where

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 2 \\ 6 & -3 & 1 \end{pmatrix} \quad \text{and} \quad \vec{b} = \begin{pmatrix} 6 \\ 4 + t \\ 2 \end{pmatrix}.$$

Following the notation used in class, define

$$A_1 = \begin{pmatrix} 6 & 2 & 3 \\ 4 + t & 5 & 2 \\ 2 & -3 & 1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 6 & 3 \\ 2 & 4 + t & 2 \\ 6 & 2 & 1 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 1 & 2 & 6 \\ 2 & 5 & 4 + t \\ 6 & -3 & 2 \end{pmatrix}.$$

Using determinant formulas, we get

$$\det(A) = -77, \quad \det(A_1) = -11t, \quad \det(A_2) = -17t, \quad \det(A_3) = -154 + 15t.$$

Since $\det(A) \neq 0$, the system has a unique solution for every $t \in \mathbb{R}$. By Cramer's rule,

$$x = \frac{\det(A_1)}{\det(A)} = \frac{-11t}{-77} = \frac{t}{7}, \quad y = \frac{\det(A_2)}{\det(A)} = \frac{-17t}{-77} = \frac{17t}{77}, \quad z = \frac{\det(A_3)}{\det(A)} = \frac{-154 + 15t}{-77} = 2 - \frac{15t}{77}.$$

Hence the solution is

$$\begin{pmatrix} \frac{t}{7} \\ \frac{17t}{77} \\ 2 - \frac{15t}{77} \end{pmatrix}.$$

- (b) Consider the linear system in the variables x, y, z given by

$$\begin{cases} 5x - y + 2z = 3 \\ x + 4y - z = 1 \end{cases}$$

Find the general solution using Cramer's rule.

Hint. For each $z \in \mathbb{R}$, compute x and y using Cramer's rule for the 2×2 system in the variables x and y .

Solution: Let $z \in \mathbb{R}$. Rearrange the system as a 2×2 system in the variables x, y :

$$\begin{cases} 5x & - & y & = & 3 - 2z \\ x & + & 4y & = & 1 + z \end{cases}$$

Write it as $A\vec{x} = \vec{b}$ with

$$A = \begin{pmatrix} 5 & -1 \\ 1 & 4 \end{pmatrix} \quad \text{and} \quad \vec{b} = \begin{pmatrix} 3 - 2z \\ 1 + z \end{pmatrix}.$$

Then

$$A_1 = \begin{pmatrix} 3 - 2z & -1 \\ 1 + z & 4 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 5 & 3 - 2z \\ 1 & 1 + z \end{pmatrix}.$$

Compute determinants:

$$\det(A) = 5 \cdot 4 - (-1) \cdot 1 = 21,$$

$$\det(A_1) = 4(3 - 2z) - (-1)(1 + z) = 13 - 7z, \quad \det(A_2) = 5(1 + z) - (3 - 2z) = 2 + 7z.$$

By Cramer's rule,

$$x = \frac{\det(A_1)}{\det(A)} = \frac{13 - 7z}{21}, \quad y = \frac{\det(A_2)}{\det(A)} = \frac{2 + 7z}{21}.$$

Therefore the general solution is

$$\left\{ \begin{pmatrix} \frac{13-7z}{21} \\ \frac{2+7z}{21} \\ z \end{pmatrix} : z \in \mathbb{R} \right\}.$$

2. (a) Consider the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{pmatrix}.$$

Compute $\text{adj}(A)$.

Hint. Use the definition.

Solution:

Let $M_{ij}^{(A)}$ be the (i, j) minor of A . Then, by definition,

$$\text{adj}(A) = \begin{pmatrix} +M_{11}^{(A)} & -M_{21}^{(A)} & +M_{31}^{(A)} \\ -M_{12}^{(A)} & +M_{22}^{(A)} & -M_{32}^{(A)} \\ +M_{13}^{(A)} & -M_{23}^{(A)} & +M_{33}^{(A)} \end{pmatrix} = \begin{pmatrix} -2 & -2 & 2 \\ 1 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}.$$

(b) Let $A \in M_3(\mathbb{R})$ be an invertible matrix such that

$$A^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ 2 & 0 & 0 \\ 0 & 3 & 1 \end{pmatrix}.$$

Compute $\text{adj}(A)$.

Hint. First explain why such a matrix A exists.

Solution: First, note that

$$\det(A^{-1})^{\text{first row}} = -\det \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = -2 \neq 0,$$

so A^{-1} is invertible and therefore such a matrix A exists. We have seen in class that $A^{-1} = \frac{\text{adj}(A)}{\det(A)}$, and therefore $\text{adj}(A) = \det(A)A^{-1}$. Since $\det(A) = \frac{1}{\det(A^{-1})} = -\frac{1}{2}$, we conclude that

$$\text{adj}(A) = \det(A)A^{-1} = \left(-\frac{1}{2}\right) \begin{pmatrix} 0 & 1 & 0 \\ 2 & 0 & 0 \\ 0 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{1}{2} & 0 \\ -1 & 0 & 0 \\ 0 & -\frac{3}{2} & -\frac{1}{2} \end{pmatrix}.$$

3. Let $A, B \in M_n(\mathbb{F})$ be two invertible matrices.

(a) Prove that

$$\text{adj}(A^{-1}) = \frac{A}{\det(A)}.$$

Solution: Since A is invertible, we have

$$A^{-1} = \frac{\text{adj}(A)}{\det(A)},$$

and therefore $\text{adj}(A) = \det(A)A^{-1}$. Applying this identity to the invertible matrix A^{-1} gives

$$\text{adj}(A^{-1}) = \det(A^{-1}) \cdot (A^{-1})^{-1} = \frac{1}{\det(A)} \cdot A = \frac{A}{\det(A)}.$$

(b) Prove that $\text{adj}(A) \cdot \text{adj}(B) = \text{adj}(BA)$.

Solution: Since A and B are invertible, we have

$$\text{adj}(A) = \det(A)A^{-1}, \quad \text{adj}(B) = \det(B)B^{-1}.$$

Also, BA is invertible and

$$\text{adj}(BA) = \det(BA)(BA)^{-1}.$$

Therefore,

$$\text{adj}(A) \text{adj}(B) = \det(A) \det(B) A^{-1} B^{-1} = \det(BA) (BA)^{-1} = \text{adj}(BA).$$

(c) Suppose that $B^2 = I_n$ and that $\det(B) = 1_{\mathbb{F}}$. Prove that $\text{adj}(A) \cdot AB = \text{adj}(AB) \cdot A$.

Solution:

We know that $\text{adj}(AB) = \text{adj}(B) \cdot \text{adj}(A)$ and that $\text{adj}(A) \cdot A = \det(A) \cdot I_n$. Therefore

$$\begin{aligned} \text{adj}(A) \cdot AB &\stackrel{B^2=I_n \Rightarrow B^{-1}=B}{=} \text{adj}(A) \cdot AB^{-1} = \text{adj}(A) \cdot A \cdot \frac{\text{adj}(B)}{\det(B)} \stackrel{\det(B)=1_F}{=} \text{adj}(A) \cdot A \cdot \text{adj}(B) \\ &= \det(A) \cdot I_n \cdot \text{adj}(B) = \text{adj}(B) \cdot \det(A) \cdot I_n = \text{adj}(B) \text{adj}(A) \cdot A = \text{adj}(AB) \cdot A \end{aligned}$$

4. Student X observed that in Q1(b) one can also treat x as a free parameter and then solve for y, z (as functions of x) using Cramer's rule on a 2×2 system in the variables y and z . She suspected that this "change of free parameter" is governed by a general determinant identity.

Let $\vec{c}_1, \vec{c}_2, \vec{c}_3, \vec{b} \in F^2$. Define

$$A = \begin{pmatrix} \vec{c}_1 & \vec{c}_2 \end{pmatrix}, \quad A_2 = \begin{pmatrix} \vec{c}_1 & \vec{b} - z \cdot \vec{c}_3 \end{pmatrix}, \quad \forall z \in F,$$

and also

$$B = \begin{pmatrix} \vec{c}_2 & \vec{c}_3 \end{pmatrix}, \quad B_2 = \begin{pmatrix} \vec{b} - x \cdot \vec{c}_1 & \vec{c}_3 \end{pmatrix}, \quad \forall x \in F.$$

Assume that $\det(A) \neq 0_F$ and $\det(B) \neq 0_F$. Prove that

$$\left\{ \frac{\det(A_2)}{\det(A)} : z \in F \right\} = \left\{ \frac{\det(B_2)}{\det(B)} : x \in F \right\}.$$

Hint. Assume first that $\vec{c}_1$ and $\vec{c}_3$ are linearly independent. **Solution:**

Suppose first that $\vec{c}_1$ and $\vec{c}_3$ are linearly independent. Define

$$a = \frac{\det \begin{pmatrix} \vec{c}_1 & \vec{b} \end{pmatrix}}{\det \begin{pmatrix} \vec{c}_1 & \vec{c}_2 \end{pmatrix}}, \quad b = \frac{\det \begin{pmatrix} \vec{c}_1 & \vec{c}_3 \end{pmatrix}}{\det \begin{pmatrix} \vec{c}_1 & \vec{c}_2 \end{pmatrix}}.$$

Since $\vec{c}_1$ and $\vec{c}_3$ are linearly independent, we have $b \neq 0_F$. Consider the set

$$R = \{a - bz : z \in F\}.$$

We claim that $R = F$. Indeed, given $x \in F$, choose $z = b^{-1}(a - x)$, and then $a - bz = x$.

Now compute:

$$\left\{ \frac{\det(A_2)}{\det(A)} : z \in F \right\} = \left\{ \frac{\det \begin{pmatrix} \vec{c}_1 & \vec{b} - z \cdot \vec{c}_3 \end{pmatrix}}{\det \begin{pmatrix} \vec{c}_1 & \vec{c}_2 \end{pmatrix}} : z \in F \right\} = \{a - bz : z \in F\} = F.$$

Similarly,

$$\left\{ \frac{\det(B_2)}{\det(B)} : x \in F \right\} = \left\{ \frac{\det \begin{pmatrix} \vec{b} - x \cdot \vec{c}_1 & \vec{c}_3 \end{pmatrix}}{\det \begin{pmatrix} \vec{c}_2 & \vec{c}_3 \end{pmatrix}} : x \in F \right\} = F.$$

Hence the two sets are equal.

Assume now that $\vec{c}_1$ and $\vec{c}_3$ are linearly dependent. Then $\det \begin{pmatrix} \vec{c}_1 & \vec{c}_3 \end{pmatrix} = 0_F$. Since $\det(B) \neq 0_F$, we have $\vec{c}_3 \neq 0$, hence there exists

$\alpha \in F$ such that $\vec{c}_1 = \alpha \vec{c}_3$. Then

$$\left\{ \frac{\det(A_2)}{\det(A)} : z \in F \right\} = \left\{ \frac{\det(\vec{c}_1 \quad \vec{b})}{\det(\vec{c}_1 \quad \vec{c}_2)} \right\} = \left\{ \frac{\det(\vec{b} \quad \vec{c}_3)}{\det(\vec{c}_2 \quad \vec{c}_3)} \right\} = \left\{ \frac{\det(B_2)}{\det(B)} : x \in F \right\}.$$

5. Let $A \in M_n(F)$.

(a) Suppose A is invertible. Prove that if $\text{adj}(A)$ is symmetric then A is symmetric.

Solution: As A is invertible then $A^{-1} = \frac{\text{adj}(A)}{\det(A)}$. Inverting both sides of the equation, we obtain

$$A = \left(\frac{\text{adj}(A)}{\det(A)} \right)^{-1} = \det(A) \cdot (\text{adj}(A))^{-1}.$$

It follows that

$$\begin{aligned} A^t &= [\det(A) \cdot (\text{adj}(A))^{-1}]^t \stackrel{(\alpha A)^t = \alpha \cdot A^t}{=} \det(A) [(\text{adj}(A))^{-1}]^t \\ &\stackrel{(A^{-1})^t = (A^t)^{-1}}{=} \det(A) \cdot [(\text{adj}(A))^t]^{-1} \stackrel{\text{symmetric}}{=} \det(A) \cdot (\text{adj}(A))^{-1} = A \end{aligned}$$

so A is symmetric.

(b) Give an example where $\text{adj}(A)$ is symmetric but A is not.

Solution: Consider the matrix defined by

$$A = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Clearly, A is not symmetric, since

$$A^t = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \neq A$$

By definition, $\forall 1 \leq i, j \leq 3$ we have

$$[\text{adj}(A)]_{ij} = (-1)^{i+j} M_{ji}^{(A)},$$

where $M_{ji}^{(A)}$ is the ji 'th minor of A , namely, the determinant of the matrix obtained from A by removing the j 'th row and the i 'th column. Since A has two different rows of zeros, this matrix has at least one row of zeros, and therefore $M_{ji}^{(A)} = 0, \forall 1 \leq i, j \leq 3$.

It follows that

$$\text{adj}(A) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

so $\text{adj}(A)$ is symmetric but A is not.

6. Let $A, B \in M_n(F)$. Prove or disprove each of the following statements:

(a) Let $\alpha_1, \alpha_2, \alpha_3 \in F$. Then

$$\text{adj}(\text{diag}(\alpha_1, \alpha_2, \alpha_3)) = \text{diag}(\alpha_2\alpha_3, \alpha_1\alpha_3, \alpha_1\alpha_2).$$

Solution: True. Let

$$A = \text{diag}(\alpha_1, \alpha_2, \alpha_3) = \begin{pmatrix} \alpha_1 & 0_F & 0_F \\ 0_F & \alpha_2 & 0_F \\ 0_F & 0_F & \alpha_3 \end{pmatrix}.$$

Compute

$$M_{11}^{(A)} = \det \begin{pmatrix} \alpha_2 & 0_F \\ 0_F & \alpha_3 \end{pmatrix} = \alpha_2\alpha_3, \quad M_{22}^{(A)} = \det \begin{pmatrix} \alpha_1 & 0_F \\ 0_F & \alpha_3 \end{pmatrix} = \alpha_1\alpha_3, \quad M_{33}^{(A)} = \det \begin{pmatrix} \alpha_1 & 0_F \\ 0_F & \alpha_2 \end{pmatrix} = \alpha_1\alpha_2.$$

If $i \neq j$, then deleting row i and column j produces a 2×2 matrix with a zero row or a zero column, hence $M_{ij}^{(A)} = 0_F$. Therefore

$$\text{adj}(A) = \text{diag}(\alpha_2\alpha_3, \alpha_1\alpha_3, \alpha_1\alpha_2).$$

(b) We have $\text{adj}(-A) = -\text{adj}(A)$.

Solution: False. Choose $A = I_3 = \text{diag}(1, 1, 1)$. Then by the previous item, $\text{adj}(A) = \text{adj}(I_3) = \text{diag}(1, 1, 1) = I_3$. But $\text{adj}(-A) = \text{adj} \text{diag}(-1, -1, -1) = \text{diag}(1, 1, 1) \neq -I_3 = -\text{adj}(A)$.

(c) We have $\text{adj}(A + B) = \text{adj}(A) + \text{adj}(B)$.

Solution: False. Choose $A = \text{diag}(1, 0, 0)$ and $B = \text{diag}(0, 1, 0)$. We know that for every $\alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$ we have

$$\text{adj}(\text{diag}(\alpha_1, \alpha_2, \alpha_3)) = \text{diag}(\alpha_2\alpha_3, \alpha_1\alpha_3, \alpha_1\alpha_2).$$

It follows that

$$\text{adj}(A) = \text{adj} \text{diag}(1, 0, 0) = \text{diag}(0, 0, 0) = 0_3, \quad \text{adj}(B) = \text{adj} \text{diag}(0, 1, 0) = \text{diag}(0, 0, 0) = 0_3$$

and therefore $\text{adj}(A) + \text{adj}(B) = 0_3 + 0_3 = 0_3$, but

$$\text{adj}(A + B) = \text{adj} \text{diag}(1, 1, 0) = \text{diag}(0, 0, 1) \neq 0_3 = \text{adj}(A) + \text{adj}(B).$$

(d) Let $K \in M_{11}(\mathbb{Z}_{11})$. If K is invertible then $\det(\text{adj}(K)) \equiv 1 \pmod{11}$.

Hint. Use Fermat's little theorem.

Solution: True. Since K is invertible over $\mathbb{Z}_{11}$, we have $\det(K) \not\equiv 0 \pmod{11}$. In general,

$$\det(\text{adj}(K)) = \det(K)^{n-1},$$

so here $n = 11$ and therefore $\det(\text{adj}(K)) = \det(K)^{10}$. By Fermat's little theorem, for every nonzero $\alpha \in \mathbb{Z}_{11}$ we have $\alpha^{10} \equiv 1 \pmod{11}$. Hence $\det(\text{adj}(K)) \equiv 1 \pmod{11}$.

(e) We have

$$\sum_{j=1}^n (-1)^{1+j} [A]_{2j} M_{1j}^{(A)} = 0_F.$$

Hint. Recall a proof we saw in class.

Solution: True. Note that the sum $\sum_{j=1}^n (-1)^{1+j} [A]_{2j} M_{1j}^{(A)}$ is the determinant along the first row of the matrix obtained from A by replacing the first row with the second row (without altering the second row). That new matrix has two equal rows, and since the determinant is alternating, then

$$\sum_{j=1}^n (-1)^{1+j} [A]_{2j} M_{1j}^{(A)} = 0_F.$$